MYP4 Unit 1 Design Summative Portfolio
- The Re-Gen Bin

Product Title

Re-Gen Bin: Household Compost Bin for recycling food waste



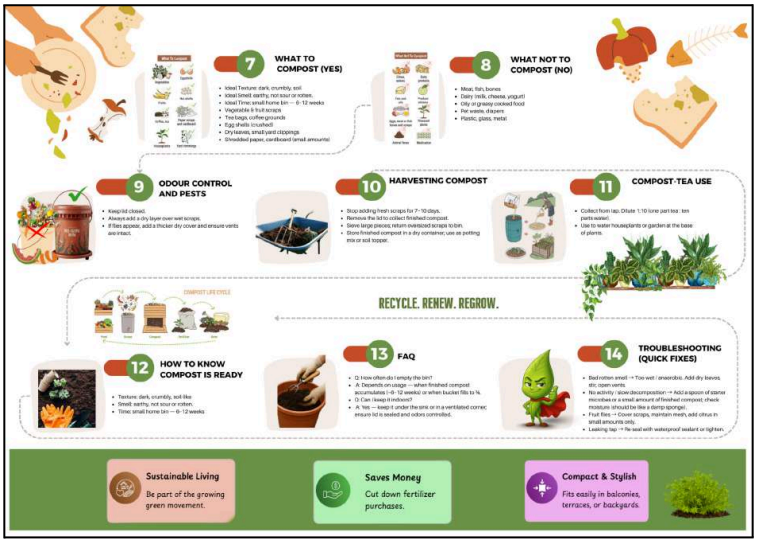




Re-Gen Bin compost bin with complete kit

(20 Ltr Bin with lid and tap, wheeled stand, 4 measuring cups, strainer, charcoal filter bag, hand gloves, gardening claw and compost matter)





User Manual

Problem Statement

Many Indian households do not practice composting, which increases the amount of food waste produced.

The Solution: The Re-Gen Bin







Target Audience

The Indian Household.

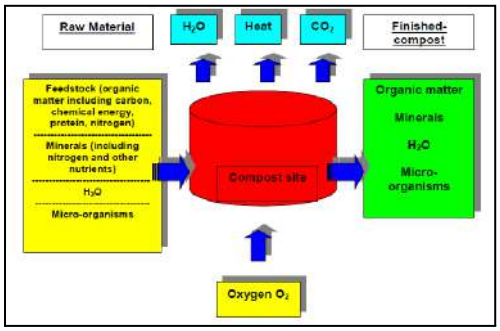
Section 1: INQUIRING & ANALYSING:

The Problem:

What is the problem?

Saying that household food waste is a gigantic problem for this country is an understatement, with an average of 55 kgs of food waste being produced annually by each of the hundred-thousands of households in India.

A very common solution for this is composting, a natural process where organic waste such as food can be put into a container called a composting bin, where microorganisms break and decompose it, while which the matter releases nutrients, transforming it into compost, a useful nutrient booster for soil and plants. Like this, food and byproducts of food that are wasted can be used to make healthy plants and more food, converting the process of food production and wastage into a sustainable and controlled loop.





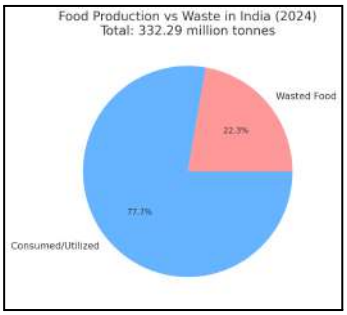

Composting, however (at least as it is done today,) has accessibility issues. Common households such as environmentally-conscious ones with a desire to turn their food waste into something useful, as well as gardening enthusiasts often face difficulty in the composting process due to the sheer complexity of how much of what ingredients to use, as well as the strong smell, which compost bins at the affordable level (which are usually between ₹200-1000) don't do much to solve.
I myself have seen examples of this:

- For example, in our school's kitchen situation, for example, when I asked them what they do to dispose of food waste, they said they simply throw it in the trash. Of course, having a compost bin in school would not be viable considering how much the compost would smell.
- I have also seen several neighbors who take interest in gardening practicing composting only briefly before quitting, owing to the complications of composting.

Why is solving this problem important?

Impact of Food Wastage:
The accessibility issues associated with composting discourage households from practicing composting. Here are some relevant numbers that demonstrate the gravity of this situation:

- According to the press information bureau of India, the country produced about **332.29 million tonnes** of food grain in 2024.
- It is estimated that **74 million tonnes** of that food is wasted. That is **22%** of food produced wasted.





- Narrowing it down further to a smaller level, **55 kg** of food is wasted every year in India per capita (per household). That represents **70%** of the food waste.
    - Though it is lower than the global average of 79 kg of wasted food per capita, considering India's population of 1.46 billion, this fact is no more very heartening. About 194 million and many more uncounted people in India don't have proper access to food, which does of course bring down the amount of food wasted by a small margin, but that is a problem of its own.
- This food waste contributes to landfills, in which methane is released, a greenhouse gas 60 times better at heat trapping than carbon dioxide, contributing to the greenhouse effect.
- India's total food waste represents **₹920 billion** of loss.

Via this data it can be seen that the problem of food wastage in India is very severe and is in desperate need for a

solution. More people practicing composting (which is possible once the accessibility problems associated with composting and composting bins are solved) will help bring down these terrifying numbers.
This can be part of an effort along with many other products and solutions to bring about a sustainable world where society, the environment and the economy are stable and ever-progressing. While one cannot attribute a single product to such a great change, controlling and managing our food waste should be the first step in bringing about such a world.

Composting vs Alternatives:

- **Composting and Food Banks**
  - Donating food waste to food banks, social welfare organisations that provide food to the underprivileged, is a commonly proposed solution. It does, however, have the limitation that while it makes sure that edible food waste is used, inedible food waste such as fruit and vegetable peels cannot be donated.
  - In composting, however, edible and inedible food waste, as well as other non-food substances such as paper and cardboard can be recycled into compost. The compost can then be added to soil to provide nutrients to plants, including edible plants, which converts the process of food production and wastage into a loop.

Who is this a problem for?

As evident from the above section, this problem affects the environment, as well as the economy. The environment is everything that is around us and the economy is representative of the product of all of our collective efforts. But the problem of food wastage threatens both of them. Therefore food wastage is a threat to the wellbeing of society.
Since here I am focusing on making a product for household use, my target audience is the **Indian Household**.

Where is the problem situated?

This problem of food waste occurs in households around the globe. In the making of my solution, I will focus particularly on the occurrence of this problem in **Indian Households**.

Final Problem Statement:

Many Indian households do not practice composting, which increases the amount of food waste produced.

Action Plan:

Research Topics

1) Justify the need for this product.
   a) Compare the amount of food produced vs food wasted.
   b) Identify the impact of food waste on various aspects. (environment, society, etc.)
   c) Identify why households do not participate in composting.
2) Identify what are important features for a composting bin to have.
   a) Interview people who have practiced composting in the household about the difficulties they have faced.
   b) Find information on the same topic on the internet.
   c) List, using the data obtained via above step, the features that a compost bin should have
3) Analyse existing compost bins.
   a) Find the top-rated compost bins from Amazon.in.
   b) Compare specifics of said compost bins based on features listed in topic no.4.
   c) Conduct a SWOT analysis of each of the compost bins.

Using the information gathered from the above steps, I will create a design brief.

Sources
Sources for Secondary Research
Articles from government websites
Articles from websites that specialise in science
Articles from websites that specialise in gardening
Articles from websites that specialise in environmental awareness
Amazon.in
Sources for Primary Research
Family members, relatives and neighbors who practice or have practiced composting

Step	Type	Purpose	Timeline	Priority
Research				
Research on the severity of the problem of food wastage. (Presented in the previous section.)	Secondary, qualitative, quantitative research	To create a problem statement and justify why the problem is relevant and needs a solution.	04/08/2025	1st priority, since it forms the base of my research and proves the project's relevance.
Research on why composting is important. (Presented in the previous section.)	Secondary, qualitative research	To create a problem statement and justify why my to-be-made product is relevant.	05/08/2025	1st priority, since it forms the base of my research and proves the project's relevance.
Conducting an interview with 2 people with varying experience in gardening and composting.	Primary, qualitative research	To understand people's problems with composting.	06/08/2025	2nd priority, since it will help me gain further knowledge about the problem and will form the basis for the 3rd priority step.
Conducting research on the problems that come with composting and composting bins.	Secondary, qualitative research	To understand people's problems with composting.	06/08/2025	2nd priority, since it will help me gain further knowledge about the problem and will form the basis for the 3rd priority step.
Analysis				
Analysing the features of 4 different existing solutions.	Secondary, qualitative research, Analysis	To look at existing products to understand the gaps in the solutions.	07/08/2025 - 08/08/2025	3rd priority, since analysis will be dependent on findings of the 2nd priority step.
Finally, based on all this research and analysis about problems and solutions, I will create a design brief.				

Research Summary:

Compost bin requirements

Based on primary research, I have concluded that there are 6 key features of an ideal compost bin. Note that this is not a final design brief. Rather, these points lay out the base for my analysis of existing products, upon which I will create my final design brief.

- A compost bin should be compact, and in such a shape that it takes up less horizontal space but still is able to fit an adequate amount of matter to facilitate proper decomposition.
- A compost bin should have some sort of feature that blocks the odour.
- A compost bin should have good ventilation.
- A compost bin should have some kind of indication as to how much of what kind of matter is to be put in.
- A compost bin should provide easy access to the compost tea.
- A compost bin should be handlable so that the stirring is easier.

Justification of analysis method

Here I will justify why I am using the analysis method I am using.

My Method:
I have identified, using my primary research, a few different attributes that I can analyse the existing products with. Those are:

- Size
- Capacity
- Material
- Cost
- Easy-handling
- Ingredient measurement method
- Compost-tea access
- Odor control
- Airflow

My method is somewhat similar to ACCESSFM, which is also a method where I would analyse different products based on a few attributes, those being:

- Aesthetics
- Cost
- Customer
- Environmental impact
- Size
- Safety
- Function
- Material

Such a method can be very efficient, especially when you want to analyse products based on the different ways they accomplish the same goals. Hence I wanted to use a method that is derivative of ACCESSFM, but modified for reasons stated below.

Why not ACCESSFM:
I have not used ACCESSFM itself, because I felt that the very specific product range I was working with made some of the ACCESSFM points irrelevant or too broad.

- My product is going to prioritize function, as in it has to have certain specific functions. Hence, I could not analyse existing products based on the broad point ‘function.’ Rather, I divided the ‘function’ point into the actual

different functions I think it is important for a compost bin to have.

- ➢ 'Aesthetics' isn't much of a priority for me in my design process here, since I am to make a product that focuses on functionality. Therefore, I felt aesthetics was not a relevant analysis point for my chosen existing products.
- ➢ The 'customer' point was not relevant for my analysis either, since I was looking specifically at compost bins for household use, which would all have a common customer, the household, anyway.

I wanted to use a method like ACCESSFM, where I would analyse different products on the same points, but would have to add more analysis points and remove some in order to fit my product type. Hence, I decided to modify ACCESSFM using the analysis points I derived from my research.

Existing solutions:

Out of the below solutions, I will chose four of the best products to analyse.

Product name	Product image	Chosen for analysis	Reason
Trustbasket indoor composter		**Yes**	Since this was the highest rated and most popular compost bin on amazon.in (probably due to its comparatively cheaper price and features such as a closeable vent and a tap to access liquid compost, or compost-tea), I found myself obliged to consider it, since its high popularity told me that there were some aspects of this product I may get inspiration from for my own solution.
AKXY home composting machine		**No**	This product is very similar to the sixth product seen in this table, the Marcytop dual-chambered composting unit, both having the idea of turning the compost bin into a spinning barrel to ease the process of stirring. I chose the other one over this one for two reasons. The first being that this composter is completely filled with holes on two sides, which do not help with odor reduction. The second reason is mentioned in the section for the Marcytop dual-chambered composting unit.
Green bin outdoor composter		**No**	This product is a very large compost bin, too big to keep conveniently in a house, and hence is not suitable for my target audience of households.
Waqif kitchen countertop compost bin		**No**	This product is absolutely tiny, as can be seen in the picture, and is meant for *extremely* small-scale composting and wouldn't even be able to accommodate a day's worth of vegetable peels in an Indian kitchen.. The product I aim to make is a full-fledged composter meant to take in a large amount of material.

Ez4garden garden leaf waste bag		**No**	This product is made of fabric. Fabric is not the ideal material for controlling odors, since it traps molecules and bacteria that may cause odor. Therefore it is not a solution my solution can take any inspiration from in some aspects.
Marcytop dual-chambered composting unit		**Yes**	This product is very similar to the AKXY home composting machine, the second product presented in this table, both having the idea of turning the compost bin into a spinning barrel to ease the process of stirring. I chose this one over the other since it offers the advantage of having two split compartments so that two different compost cycles can go on at the same time, so that you can remove compost from the section that gets ready first without having to wait for everything to get ready.
Kibaga household composting bin		**Yes**	In a very unique way, this product is made of steel and charcoal, both materials that are good at preventing odors. Therefore this is a product I may borrow aspects from.
Daily dump aerobic Prithvi Khamb		**Yes**	This product is made of clay, which has odor-controlling properties due to how it contains minerals which can absorb or neutralise odor-causing molecules. Additionally, it has a separable lower chamber which can make it very easy to access the liquid compost, or compost-tea.
Velura inground worm composter		**No**	This product is made of plastic. Plastic, due to its porous structure through which molecules can pass, is not a good material for the controlling of odors. Therefore it is not a product my product can take any inspiration from in some aspects.
Dostbin smart automatic composter		**No**	This product is a large and expensive (₹34,500) electronic product. Though this product can compost much quicker due to being electronic, this product is not ideal for my target audience of households due to its great price and size, and is rather ideal for small businesses or farmers, who can actually pay that amount since they will be making a profit out of it.

Other solutions

Best existing solutions:
For this section, I have chosen the 4 best compost bins from amazon.in.

Attributes	**Kibaga household composting bin** 	**Marcytop dual-chambered composting unit** 	**Daily dump aerobic Prithvi Khamb** 	**Trustbasket indoor composter** 
Size	17.8L x 17.8W x 25.9H Centimeters	74.9L x 67.8W x 95.3H Centimeters	38L x 38W x 85H Centimeters	31.8L x 31.8W x 29.2H Centimeters
Capacity	3.99 kg	170.55 kg	60 kg	14 kg
Material	Stainless steel, charcoal	Plastic	Clay	Plastic
Cost	₹9719	₹11560.40	₹3500	₹999
Easy-handling	It has a handle, making it easy to transport and to stabilize it while stirring.	It has dual compartments that can be rotated on a rod, negating the need for manual stirring	Due to the combination of being made of clay and not having a proper handle, it may be breakable.	It has a handle, making it easy to transport and to stabilize it while stirring.
Ingredient measurement method	None	None	None	None
Compost-tea access	None	None	Separable ‘bowl’ section to accumulate compost tea.	Tap at the bottom of the bin.
Odor control	Odor control via charcoal and steel structure neutralises odors.	Because the compost can be stirred without need for opening, the smell can be avoided.	Clay absorbs and neutralises odors	None
Airflow	Has a few holes for ventilation on the lid.	None	Has holes for ventilation throughout.	Has a closable vent at the lid.

In the above table I have noted how four different existing products accomplish various goals. Taking some inspiration from these, I will create a design brief.

Design Brief:

The following restates and summarises points from various parts of the document in one place as a design brief.

Problem statement

Many Indian households do not practice composting, which increases the amount of food waste produced.

Target audience

The Indian household

Problem details

Indian households, including those of gardening enthusiasts, are discouraged from composting due to accessibility problems such as:

- The complexity of the process
- The odor the compost gives off.
    - This is usually ignored by composting bins in favour of vents and holes for airflow. If a compost bin is completely closed, there will be less of a smell, but the process would fail due to the lack of airflow.
- Space constrains

Problem severity

- **332.29 million tonnes** of food grain produced in India in 2024.
- **74 million tonnes** (**22%**) of food wasted.
- **55 kg** (**70%**) food waste per capita.
- This food waste contributes to landfills, in which methane is released, a greenhouse gas 60 times better at heat trapping than carbon dioxide, contributing to the greenhouse effect.
- India's total food waste represents **₹920 billion** of loss.

Solution goal:

To create a compost bin that solves the accessibility problems that discourage households from composting, which will encourage more people to practice composting and hence reduce the household food waste produced.

Requirements for solution

The compost bin good compost bin should:

- have some sort of feature that blocks the odour.
- have good ventilation.
- have some kind of indication as to the correct measurement of the correct kind of matter to be put in.
- be compact, and in such a shape that it takes up less horizontal space but still is able to fit an adequate amount of matter to facilitate proper decomposition.
- provide easy access to the compost tea.
- be handlable so that the stirring is easier.

Conclusion

In design, we aim to create products to bring about a sustainable world which is ever-evolving and ever-improving. It is small inventions like this **ReGen Bin** that work together to bring this reality into being. Throughout this unit, I will consider my target audience's problems and create a solution as another step towards this future.

Background Research

This section contains some of the more extensive research, only the summaries of which are presented on the main document. Though it is background research, it is included here for the sake of keeping all the research in one place where it can be easily navigated through.

Interviews

Interview with Kavitta Shirke, 44, Thane

1) Do you or did you ever practice composting? Since how long do you compost?

Yes, 6-8 months.

1) What was the procedure you used for composting?

When I did composting for the first time, I used a small waste paint bucket and inside it I put soil and fruit and vegetable peels in alternate layers and kept the compost bin in the backyard and let the process take over. Later I changed the process.

2) Why did you use this procedure?

Ultimately, the initial procedure proved incorrect. That was because I thought composting to be a simple process and began without finding the complete information. What is worse, when I did some research, I found that there was varying information all over the internet about the exact, specific process and ratios.

3) Were you able to compost successfully?

The initial method was not eventually successful. I never stirred the compost, I had added no holes or ventilation points to the bucket, and so one day the bucket just exploded. I later found out from the gardener that my homemade compost bin should have had some kind of ventilation, had to be stirred constantly and rather than just normal soil I had to use cocopeat.

4) Why was it successful or unsuccessful?

Initially it was unsuccessful because of there being no ventilation to compost. When I redesigned the bin by adding holes to it and started adding cocopeat to it, the compost came out well.

5) Based on this, what features do you think a good compost bin should have to solve these problems?

Based on this, I can tell that a compost bin should have good ventilation, though such that it does not smell too much.

Interview with Ujwala Pise, 53, Kandivali

1) Do you or did you ever practice composting? Since how long do you compost?

Yes, 9-10 months.

1) What was the procedure you used for composting?

I used a large plastic container kept in the balcony, inside which I added kitchen waste such as wasted food and fruit and vegetable peels chopped into small pieces, soil and cocopeat.

2) Why did you use this procedure?

I did it based on what I saw a friend do on a Facebook group.

3) Were you able to compost successfully?

The compost was coming out well, however I did eventually stop composting.

4) Why was it successful or unsuccessful?

Yes, while the compost was coming out well, I discontinued shortly after, the reason being that despite the compost bin being in the balcony, the smell always seeped into the house, especially after stirring it. I ended up giving away the excess compost to the society garden. Additionally, the bin was so big that it took away the entire balcony, and so that encouraged me to stop composting.

5) Based on this, what features do you think a good compost bin should have to solve these problems?

A compost bin should be space-efficient and compact and should have a way to block the smell. Additionally, with the container, I was not able to access the compost tea, the liquid compost, since it is located in the lower layers. So even easy compost tea access should be a feature in a compost bin.

From the above data we can conclude that a compost bin should have the following features. (Note that this is not the final design brief. The final design brief will be presented following the analysis of existing solutions.)

- ➢ A compost bin should be compact, and in such a shape that it takes up less horizontal space but still is able to fit an adequate amount of matter to facilitate proper decomposition.
- ➢ A compost bin should have some sort of feature that blocks the odour.
- ➢ A compost bin should have ventilation.
- ➢ A compost bin should have some kind of indication as to how much of what kind of matter is to be put in.
- ➢ A compost bin should provide easy access to the compost tea.
- ➢ A compost bin should be handlable so that the stirring is easier.

Section 2: DEVELOPING IDEAS:

Design Specifications:				
Specification name	**Specification description**	**Justification**	**Exact effect**	**Success criteria**
Size				
Approx. 20-30L x 20-30B x 30-40H centimetres	The Re-Gen Bin will be 30-40 centimetres in height, and 20-30 centimetres in length and breath.	What is approximately 20-30 x 20-30 sq. centimetres of horizontal space is not too big to use for a household compost bin, and 30-40 centimetres is not too big of a height, either, since homes already have dustbins of similar sizes. This size will be large enough to fit an adequate amount of compost and also fit itself efficiently in a house. As per Amazon.in, 15-20 litre buckets of glue, which was what I mentioned in the above section I wanted to use as the base of my product, is of said dimensions.	The Re-Gen Bin will be cylindrically shaped. From the top to bottom, it will be 30-40 centimetres tall, while its diameter when seen from top will be about 20-30 centimetres.	➢ The product being the accurate size as described in specifications .
Capacity				
15-20 litre capacity	The Re-Gen Bin will have a volume adequate to fit 20 litres of compost, the difference accounting for changes in unforeseen	As per the results of this survey I did of households in my society, the majority of households would require a bin with a capacity of 15-20 litres (see here) to contain their household's kitchen waste. In this survey, I had entered capacities for the compost bin based on common capacities for dustbins on amazon.in.	The Re-Gen Bin will be able to store approximately 20 litres of compost. This value was found by finding the volume of a cylinder of the specified size. The difference, 15-20 litres and these	➢ The product having the accurate capacity as described in the specifications .

	variables.	Additionally, in order to solve two problems with a single solution, the problem of food waste, as well as plastic waste, I intend to use common paint/glue/wall putty buckets as the base of my product. According to my mother, an interior designer by profession, said buckets are commonly available in 15-20 litre sizes, which would make them easier to obtain from scrap dealers, allowing me to use those buckets in this project.	values being slightly less than the actual amount that is found when calculating, is for taking into account variables difficult to determine at this stage, such as the thickness of the bin and the space potentially taken up by the charcoal filter and compost tea accumulation space, as well as other variables that are yet unforeseen.	
Material				
Option 1: Plastic structure	The Re-Gen Bin will be available in plastic, a material that is affordable and durable.	Since my goal with this product is to make something that encourages composting, the product being affordable is a crucial requirement, since if the product is affordable, more people will buy it. Hence, plastic is the ideal material for the product since it is durable for long-term use, similar to other durable materials such as steel, while being significantly cheaper. Plastic has a lower manufacturing cost than other durable materials such as steel and other metals due to how it has a lower, more energy-efficient extraction cost, a lower handling and shipping cost due to its light weight and less post-production processes when compared to metals, which require a lot of energy to extract, are often heavier and need additional procedures such as adding a rust-resistant layer. (New Berlin Plastics) (RSPinc) Plastic will also be more convenient than metals like steel in a household setting due to being resistant to rust, having better impact-absorption and being lighter, meaning that they will have lower maintenance and will be easier to handle, allowing for a much better user-experience. (Atlas Fibre)	The Re-Gen Bin will be made of plastic. Plastic has been chosen. Because of its affordability and durability when compared to materials like clay or terracotta, users can choose one of the 2 versions of the product depending on their specific needs and priorities.	➢ Agreement of price among users (interviewees.

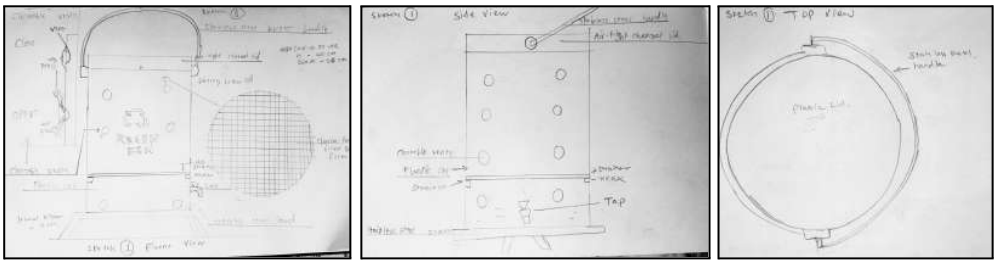
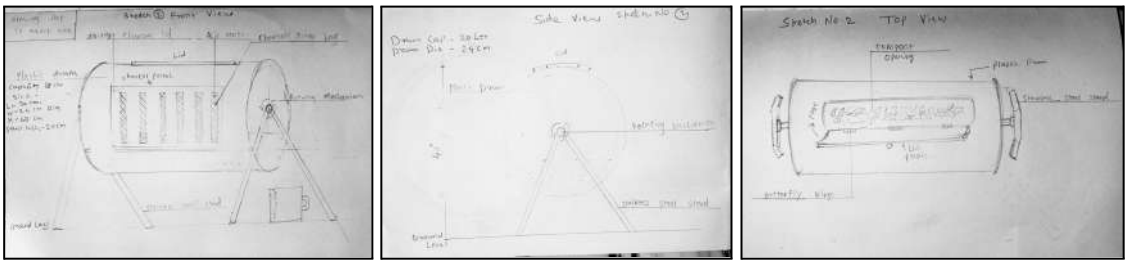
Option 2: Terracotta structure	The Re-Gen Bin will be made of terracotta, a material that has odor-reducing properties and is sustainable.	One of the most prominent accessibility problems of composting found during my primary research was the odor of compost, which needs to be blocked. Because the bin cannot be completely sealed due to the need of vents, the best way to block the odor is via an odor-proof material forming the structure.	The Re-Gen Bin will be made of terracotta. terracotta has been chosen due to its porous, layered structure and minerals can absorb or neutralise odor-causing molecules. (kloprogge)	➢ Agreement of users (interviewees) that the odor-control of this product is better than the compost bins they have used before.
Easy-handling				
Strong twin handles	The Re-Gen Bin will have strong handles in two opposite directions.	The Re-Gen Bin has a capacity of approximately 15-20 litres. Since 1 litre of compost can weigh around 0.6 kg, the Re-Gen Bin, once full, will weigh around 9-12 kg, which can prove to be difficult to carry for some people. (Roebuck) Keeping mechanical efficiency and safety in mind, strong handles to carry the product are a must.	The Re-Gen Bin will have handles in two opposite directions that can carry its weight so that the user can easily carry the bin around.	➢ Handle holding the weight of the product while full for 10 seconds without breaking anything.
Stirring claw and handgloves	The Re-Gen Bin will include a small garden claw to help with stirring and handgloves.	Stirring the compost about once a week is important to the process, since it allows oxygen to spread out and reach more microbes to provide them with the energy to speed up the process. Hence, to make it easier, considering the amount of compost that will be stored in the bin, a piece of equipment has to be included. (United States Environmental Protection Agency)	The Re-Gen Bin will include a small garden claw, 30 centimetres at length, to help the user stir the compost easily considering its volume and density.	➢ Agreement of users (interviewees) that this makes the product easier to use.
Supplementary compost matter	The Re-Gen Bin will include 1kg bags of cocopeat, neem powder, turmeric powder and microbes, each.	The reason for including these products is to reduce the complexity of the composting process, which I found to be a huge discouragement for beginners in composting during primary research. Including supplementary compost matter will ease the process for beginners.	The Re-Gen Bin will include a 1kg bag of cocopeat to assist the user with balancing the nitrogen-rich, moist content of most food waste with the dry, moisture-absorbing carbon-rich cocopeat. This, alongside neem, turmeric powder and microbes, which are key to the decomposition process, will be	➢ Agreement of users (interviewees) that this makes the product easier to use.

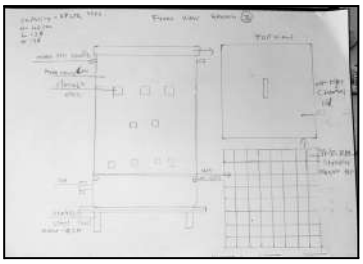
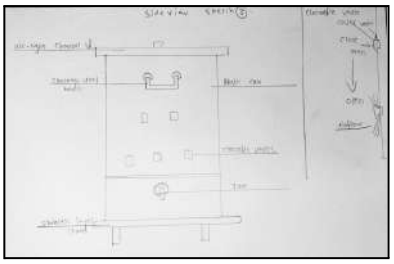
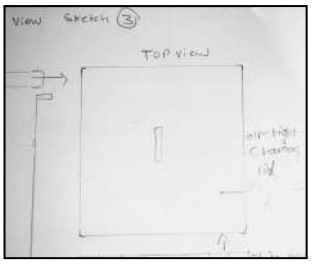
			provided in a small enough quantity for the user to add to their food waste to get started and get a hang of how the basic process works.	
Carbon steel stand	The Re-Gen Bin will have a stand to aid in the extraction of compost-tea.	The compost-tea, the liquid content of the compost, as explained in a later section, is formed at the bottom of the compost, which means that its extraction mechanism will have to be located closer to the bottom of the compost bin, which will make it difficult for the user to insert a cup and collect the liquid. Therefore, to make more clearance for the user to easily extract the compost-tea, a stand will be included. The material carbon steel has been chosen because it is more affordable than other steels due to its comparatively simple composition of easily-available raw materials, those being carbon and iron. (The Welding Institute) Despite this, due to its carbon content, it is still stronger than other metals of similar cost, such as mild steel and aluminium due to its carbon content. (Brikov, Michail et al.)	The Re-Gen Bin will include a stand at least 4-6 inches tall, which will be strong enough to hold up the compost bin so that the user will offer enough clearance to easily use the compost-tea extracting mechanism located at the lower portion of the compost bin.	➢ The stand being able to hold the compost bin at full capacity for 3 days ➢ The stand being stable with the compost bin kept on it.
Odor control				
Odor-controlling activated charcoal bag	The Re-Gen Bin will include an activated charcoal bag to further eliminate the odor.	One of the most prominent accessibility problems of composting I identified during my research was the odor of compost, which needs to be blocked.	The Re-Gen Bin will include a mesh bag filled with charcoal. This mesh will be able to be kept directly below the lid and at the inner walls of the bin. Using the molecule-trapping, porous structure of charcoal, it will help further eliminate the odor. (Mohan)	➢ Agreement of users (interviewees) that the Re-Gen Bin releases less foul odor than other compost bins they have with in the past.
Ingredient measurement method				
Measuring cups	The ReGen Bin will include measuring cups	From the interviews conducted earlier, I found that one of the problems discouraging people from	The Re-Gen Bin will include a range of cups, each assigned	➢ Agreement of users (interviewees)

	included for each type of compost matter, such as carbon-rich and nitrogen-rich matter and cocopeat, with the right amount of each material by its size.	practicing composting was the complexity of the process, specifically in terms of the amounts and ratios of different ingredients. Therefore, this feature is necessary.	a different type of material used in compost, such as carbon-rich and nitrogen-rich matter and cocopeat. Each of them will be sized based on a scientifically-correct ratio of said type of matter in comparison to the other cups. It will also have a diagram on it showing in which layers of the compost its respective matter should be added.	that the measuring cup system makes the process of composting easier for them to manage than it was before.
Compost-tea access				
Compost-tea access tap	The Re-Gen Bin will have a tap for accessing compost-tea (liquid compost).	Looking at the different existing compost bins, I observed that very few of them allow the user to access the compost-tea, which has its own separate uses and none provide the user with a way to more easily get its ratio with water right. Therefore, this is a gap in the solutions that my product should solve.	The Re-Gen Bin will have a tap attached to its lower part, since that is where it accumulates.	➢ The smooth opening and closing of the tap. ➢ The tap being able to release compost-tea. ➢ The tap not leaking.
Water-dilution measurement	The Re-Gen Bin will include a pair of bowls with identical scales printed on both to make it easier for the user to determine the amount of water necessary to dilute the compost-tea.	The compost-tea that is collected from the compost often contains harmful pathogens and anaerobic microbes that are not good for plants. Hence, the liquid content is supposed to be processed further before use by diluting it with 1:10 of water. Hence there needs to be a way for the user to know how to properly dilute it. Therefore this feature is necessary. (Yin et al. 1, 6)	The Re-Gen Bin will include a pair of identical bowls with a scale printed onto them, representing the volume of the liquid content inside. Seeing how much compost-tea has been collected in one bowl, the user can fill the necessary amount of water based on the 1:10 ratio into the other bowl and then use it to dilute the compost-tea and make it ready for use.	➢ A plant treated with compost-tea diluted this way having a better growth-rate than a plant of the same species and age kept under the same conditions but treated directly with undiluted compost-tea under the span of 1 week.

Airflow				
Ventilation	The Re-Gen Bin will have holes for ventilation on its sides and on its lid.	Composting is an aerobic process. It depends on microorganisms to break down matter into compost and oxygen is a crucial energy source for them. Therefore ventilation is crucial to any compost bin.	The Re-Gen Bin will have holes along its sides, as well as along its lid. These holes will allow oxygen to flow into the bin in order to provide energy to the microorganisms and boost the process.	➢ The compost being in its ideal state, that is with a black-brown colour, wet and crumbly texture, earthly smell and 70-80% of the volume of the matter converted to compost by the end of 4 months. (Environmental Protection Agency)

Sketches

Sketch No.	Sketch Title	Sketch
Note: These sketches were made when I planned to use stainless steel for many metal parts of the product. After further research on which metal would be ideal for my product, I decided to use carbon steel instead, as mentioned in the Design Specifications.		
1	**Cylindrical Design**	
2	**Horizontally-Rotating Design**	

3	**Cuboid Design**	  

I asked Mrs. Kavitta Shirke, Mr. Vikas Shirke and Mr. Atharva Pawar, people fitting into my target audience of Indian Households if these sketches are clear to them with only the context of it being a compost bin. From their responses I can conclude that the sketches look clear, though a few of them look a bit messy.

Alignment of the three Sketches with the Design Specifications

Specification	**Sketch 1**	**Sketch 2**	**Sketch 3**
Aprox. 15-20 litre capacity	✅	✅	✅
Approx. 20-30L x 20-30B x 30-40H centimetres	✅	✅	✅
Plastic structure	✅	✅	✅
Carbon steel handles	✅	N/A	✅
Stirring claw and handgloves	✅	✅	✅
Supplementary compost matter	✅	✅	✅
Stand	✅	✅	✅
Odor-controlling activated charcoal bag	✅	✅	✅
Measuring cups	✅	✅	✅
Compost-tea access tap	✅	❌	✅
Water-dilution measurement	✅	✅	✅
Ventilation	✅	✅	✅

Choosing a Final Design

Here I will choose the final design out of the three sketches via the process of elimination.

Elimination 1: Sketch 2

Though, due to the spinning feature, this design seems exotic, being positioned horizontally it will take up more area, taking up a large part of the room or balcony it is placed in. This may become a discouraging factor for households to use the product. This design for my product, while very unique, is not ideal for my target audience, the household. Additionally, though this feature will make stirring the compost easier, it will make the separation and accessing compost-tea impossible, since rather than accumulating at the bottom, the compost tea will mix with the compost during

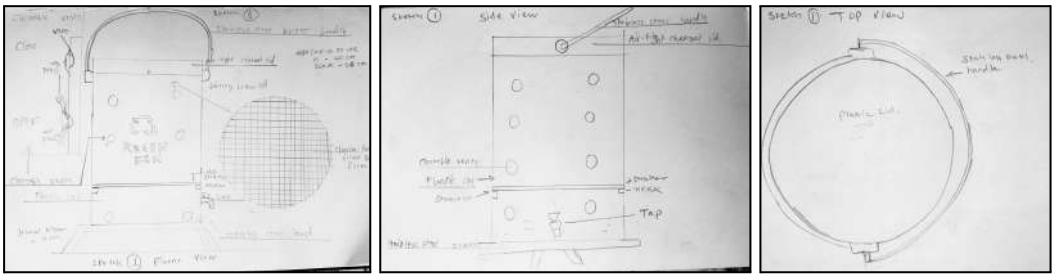
stirring. Hence Sketch 2 also skips a specification. Therefore, I eliminate Sketch 2.

Elimination 2: Sketch 3

Sketch 3 includes a charcoal lid while Sketch 1 includes a charcoal bag filter, both serving the function of odor-control. A charcoal lid will be more expensive to produce than a charcoal filter bag, since for the former, charcoal will be required to be processed into a solid form, while the latter requires charcoal powder to be filled into a mesh bag as it is. According to the The Food and Agriculture Organization of the US, to process charcoal into a solid shape, it has to be mixed with other compounds such as starch or cement, which not only requires a factory to do in mass, but would also potentially change the charcoal's porous structure, reducing or eliminating the charcoal's odor-controlling abilities. Meanwhile, the charcoal filter bag will only require charcoal to be filled into a mesh bag, which does not change or limit the charcoal's physical or chemical properties, while also not requiring as much processing. This makes the design in Sketch 1 cheaper than that of sketch 3. Additionally, as mentioned in the specifications, I intend to recycle existing paint or glue containers in order to limit the production of new plastic and in turn promote sustainability. If you search for glue buckets or paint buckets online, you are more likely to come across cylindrical ones than cuboid ones. This means that if I were to choose sketch 1, it would be more likely for me to find a bucket that can be used as the base of my product. Hence, Sketch 3 is less likely to promote sustainability than Sketch 1. (Farming and Agriculture Organization of USA)

Chosen Design: Sketch 1

The final sketch left standing is Sketch 1, for all the reasons mentioned above. Therefore, the Re-Gen Bin is finally going to be based on Sketch 1.



How does it Solve the Problem:

The problem statement was 'Many Indian households do not practice composting, which increases the amount of food waste produced' with the target audience being 'The Indian Household.' My product, the Re-Gen Bin solves this problem by creating a compost bin that eliminated common difficulties faced by households in composting, such as the odor of compost and the complex process, finding a solution to these discouraging factors of composting by implementing features such as odor-control mechanisms like a charcoal filter bag and means to make the process easier such as measuring cups. By making the practice of composting more accessible for an Indian household audience and increasing the amount of people who practice composting, this product will contribute to the reduction of food waste.

Final 3D Sketch of Chosen Design

Medium: AutoCAD

Opaque 3D Views of Re-Gen Bin

Front View



Top View



Side View



Back View



Bottom View



Video of Opaque 3D Sketch

- Video Walkthrough 1
- Video Walkthrough 2

Translucent (Internal) 3D Views of Re-Gen Bin

Note: For a clearer view of the internal components, such as the charcoal filter and wire filter mesh, I have included the following views where I have made the outer shell of the Re-Gen Bin translucent.

Video of Translucent 3D Sketch

- Video Walkthrough 1
- Video Walkthrough 2

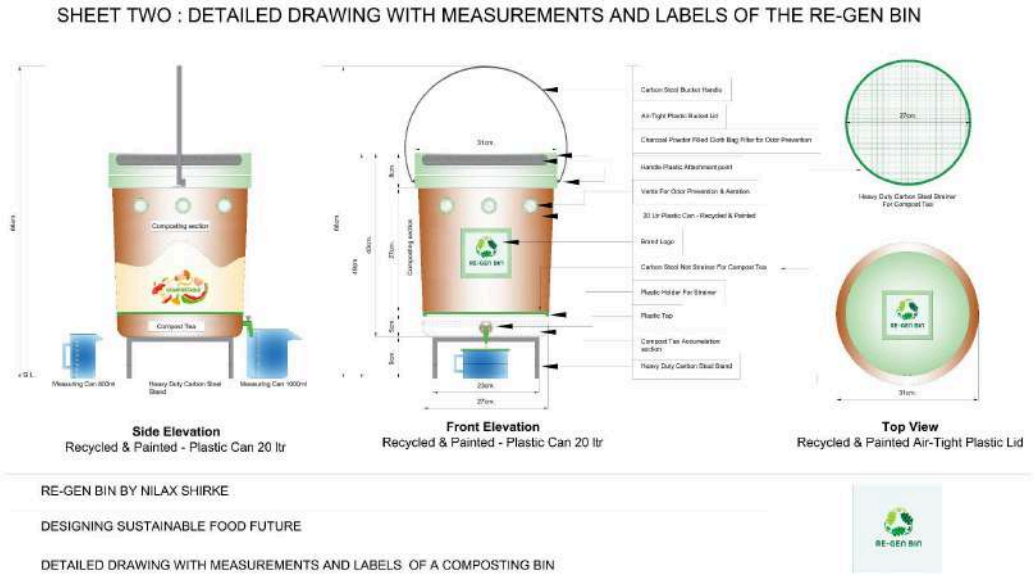


Annotations for Final 3D Sketch of Chosen Design

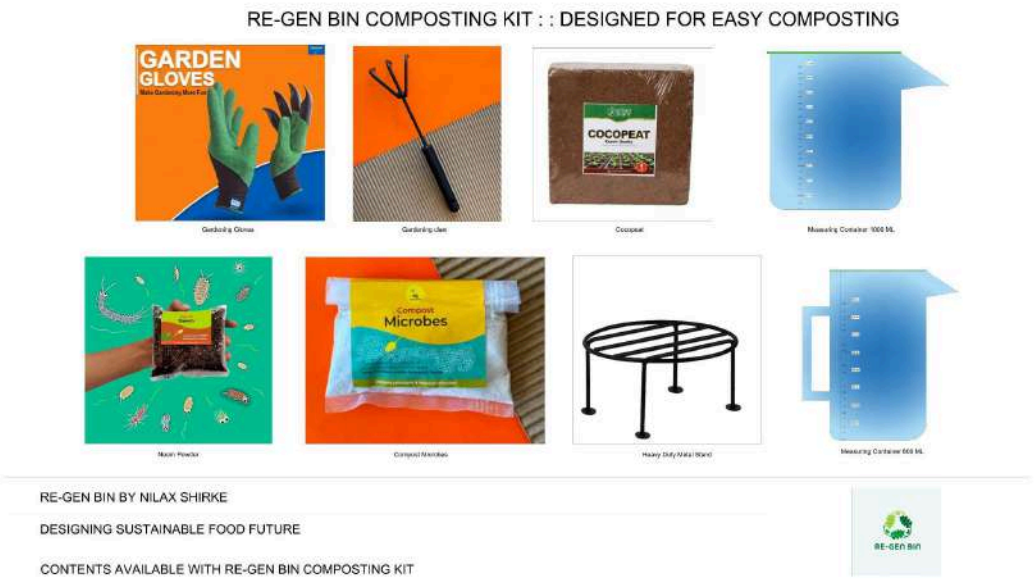
Medium: AutoCAD

Note: Since it would be impractical to add labels on a moving 3D design, I have replicated the same design with its external and internal elements in 2D to add labels on.

2D View of Re-Gen Bin (With Annotations): PDF



Re-Gen Bin Composting Kit: PDF



Alignment of the Final Sketch with the Design Specifications		
Specifications:	**Yes/No**	**How the Design follows the Specifications:**
Aprox. 15-20 litre capacity	✅	The capacity is dependent on the scale, which is correctly represented.
Approx. 20-30L x 20-30B x 30-40H centimetres	✅	The scale is correctly represented.
Plastic structure	✅	The compost bin will be made of a used plastic bucket/can.
Carbon steel handle	✅	The compost bin will have a carbon steel handle.
Stirring claw and handgloves	✅	Composting equipment will be included with the bin.
Supplementary compost matter	✅	Some ready-made compost matter will be included with the bin.
Stand	✅	The compost bin has a Metal stand to hold it up.
Odor-controlling charcoal bag	✅	A charcoal bag is attached to the bottom of the lid.
Measuring cups	✅	Measuring cups Specially-designed based on composting ratios will be included with the bin.
Compost-tea access tap	✅	There is a tap located at the lower portion to access the compost-tea.
Ventilation	✅	There are vents on all sides of the bin.

Manufacturing Process for 1 Re-Gen Bin					
Step No.	**Step:**	**Resources:**	**Approximate Estimation of Time taken:**	**Quality Checks (Success Criteria):**	**Images:**
1	Wash and dry the used plastic can that will act as the base of the product.	➢ Water ➢ Soap ➢ Cloth	10 minutes	The surface must be clean and dry and should not have moisture.	
2	Use a drill to make holes on the lid of the diameter of 1 cm on every 5 cm of the perimeter of the lid with more layers of holes being 4 cm inside the circle formed by the first layer of holes on the lid.	➢ Drill ➢ Ruler	15 minutes	The holes must be smooth and evenly spaced.	
3	Crush 1 kg of charcoal into small pieces and fill it into a bag with approximately the same dimensions. Once the bag is stitched close, attach it to the bottom of the lid.	➢ 1 kg Charcoal ➢ Mortar and Pestle ➢ Mesh bag ➢ Glue ➢ Needle ➢ Thread	1 day	The holes in the mesh bag must be small enough to not let the charcoal pieces fall out.	
4	Take a wire filter disc the same diameter as the point in the bin that is 6 cm above its lowest point and fix it onto said point.	➢ Wire filter disc ➢ Glue	5 minutes	The holes in the wire filter disc should be sized so that the compost-tea can escape into the layer below it without the solid compost passing through.	

5	Drill a hole with the diameter of 2 cm at the point directly below where the wire filter disc was attached and tightly fit the tap into it. Seal the area around it with water-proof sealant and let it dry.	➢ Tap ➢ Drill ➢ Water-pro of sealant	20 minutes	The tap must open and close smoothly and without any leaks.	
6	Use glue to fix the bin onto the heavy-duty steel stand.	➢ Heavy-dut y steel stand ➢ Glue	5 minutes	The bin and stand must be stable and must not wobble.	
7	Paint the Compost Bin in the colours shown in the sketch using oil paints and let it dry.	➢ Oil paints ➢ Paintbrush	2 hours	The paint must be evenly applied and dried.	
8	Test the final product based on the success criteria defined in the Design Specifications. If the product at this stage does not meet any particular specification or criteria, the relevant aspect of the product will be reworked.	Resources mentioned in Design Specifications.	3 weeks	All specifications must be successful.	

Feasibility

I aim to create my product by using a used industrial paint or glue bucket/can as the base and modifying it to match the product the Design Specifications describe. To accomplish the final product, **I will require many miscellaneous materials** such as charcoal, a mesh bag, a tap, a mesh disc and some kind of contraption to slightly close the vents, among other items that can be **gathered from hardware stores**. Implementing these elements into the product in the given time will take **basic crafting skills** and **time-management skills**, however evaluating if to use them or not or which versions of said items are ideal will require **judgement and analysis skills**. **All above mentioned skills, I possess**. Though most of these things I can do myself with minimal help, for certain processes, such as the making of the charcoal filter, I will need to pass the task on to an **external professional with expertise in the required skills**, such as precise sewing skills. In conclusion, in order to create this product, I will require **a combination of personal skills and external support**.

Section 3: CREATING THE SOLUTION:

Gantt Chart

https://docs.google.com/spreadsheets/d/1vw8IRmruFLbPiAqZFIRsPSxUQOBtu5LqFjcbp7coHPM/edit?usp=sharing

Justification of Timeline

- The first week is spent completely in gathering materials since buying these materials may require visiting many hardware stores, as well as trial and error regarding various materials in order to find the perfect materials and so may require a week's time.
- The second week consists of the quick tasks of assembling the materials gathered as of yet. Since the 4 smaller tasks will take a maximum of a day each, covering 4 days and considering finding a shop or organisation to help me make a custom charcoal bag may require research, 2-3 days can be reserved for this task. Hence, it makes sense for these tasks to occupy the second week.
- The third week consists of painting the Bin and fixing any unforeseen issues, as well as waiting for the charcoal filter to get completed. Painting and drying can be done easily in 2 days, while fixing unforeseen issues can go on for an unforeseen amount of time, hence having the rest of the week reserved for it. Therefore, reserving the rest of the week for these 2 tasks makes sense.
- Week 4 is the week which I estimate will be the maximum time by which I will receive the finished charcoal filter, upon the application of which, I can focus on the tests.
- Some of the tests require a lot of time, such as a composting test for several weeks. Therefore, weeks 5-7 are completely reserved for these tests. I have also placed the task of testing on every previous week since some of the tests are quick and crucial, such as checking the tap's smooth functioning once installed.

The Final Product

Main images (Front, Back, Side,Top, Bottom View, with Stand and without Stand)

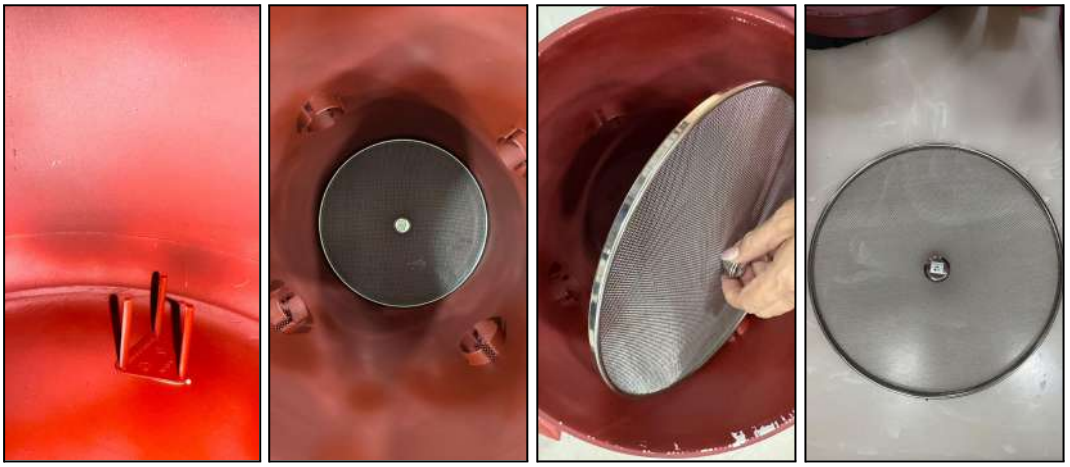




Compost Tea Extraction Tap



Compost Tea Strainer



Charcoal Filter




Ventilation



Wheeled Stand




Measuring Cups: 5 ml (Compost Tea), 50 ml (Water), 100 ml (Green Waste), 250 ml (Brown Waste)



Composting Equipment: Gardening claw & Gloves



Starter Kit: Cocopeat and Microbes



User Manual


HOW TO USE THE RE-GEN BIN
Assembly & Usage Manual
1 WHAT'S IN THE BOX
Re-Gen Bin (recycled 20 L container) with lid
Wheeled stand
Tap for compost tea extraction
Activated charcoal odor filter
Stirring claw and gardening gloves
Measuring cups for measuring compost matter
Starter compost matter (microbe mix and cocopeat)
2 RE-GEN-BIN PREPARATION
Clean bin — Wash and dry thoroughly.
Fit tap — Check if the tap is sealed and not leaking.
Place on stand — Confirm stability.
Place the strainer and keep an old cloth or paper on it.
Use measuring cups to add green waste (vegetable peels, kitchen scraps, grass clippings) and brown waste (cocopeat, dry leaves, cardboard) in a ratio of 1:2.5.
Add starter layer — 2-3 cm of dry leaves/cocopeat + a spoon of microbes. This speeds up decomposition.
3 WHERE TO KEEP IT
Place in a shaded balcony, kitchen corner, or small patio.
Keep off direct sun and away from heavy rain.
Leave 5-10 cm clearance around the bin for airflow.
4 DAILY USE – HOW TO ADD WASTE
Add kitchen scraps: veg & fruit peels, tea leaves, coffee grounds, eggshells (crushed).
Add a brown layer: a handful of dry leaves, shredded paper or cocopeat. (Always cover wet scraps with a dry layer.)
Add them according to the 1:2.5 ratio.
Close the lid firmly.
If too wet: add more dry material. If too dry: sprinkle ~50-100 ml water per 1 kg waste. (Use measuring cup.)
Step 5 WEEKLY CARE
Stir / Turn the mix with the claw every 3-4 days (2-3 minutes). This adds oxygen and speeds decomposition.
Check smell: mild earthy smell is normal. Strong rotten smell → add dry leaves and stir.
Empty compost-tea: open tap weekly into a bucket; use provided measuring cups to dilute 1:10 with water before using on plants.
Step 6 MAINTENANCE & CARE
Always wear gloves when handling compost.
Wash hands after use.
Keep away from children's reach if necessary.
Do not burn compost or add hazardous chemicals.
Unclog holes with toothpick
Re-Gen Bin is a home composting system designed to turn daily kitchen waste into healthy compost.
Made by recycling a used 20L plastic container, the bin helps families reduce food waste while creating nutrient-rich soil.


7 WHAT TO COMPOST (YES)
Ideal Texture: dark, crumbly, soil
Ideal Smell: earthy, not sour or rotten.
Ideal Time: small home bin — 8-12 weeks
Vegetable & fruit scraps
Tea bags, coffee grounds
Egg shells (crushed)
Dry leaves, small yard clippings
Shredded paper, cardboard (small amounts)
8 WHAT NOT TO COMPOST (NO)
Meat, fish, bones
Dairy (milk, cheese, yogurt)
Oily or greasy cooked food
Pet waste, diapers
Plastic, glass, metal
9 ODOUR CONTROL AND PESTS
Keep lid closed.
Always add a dry layer over wet scraps.
If flies appear, add a thicker dry cover and ensure vents are intact.
10 HARVESTING COMPOST
Stop adding fresh scraps for 7-10 days.
Remove the lid to collect finished compost.
Sieve large pieces; return oversized scraps to bin.
Store finished compost in a dry container; use as potting mix or soil topper.
11 COMPOST-TEA USE
Collect from tap. Dilute 1:10 (one part tea : ten parts water).
Use to water houseplants or garden at the base of plants.
COMPOST LIFE CYCLE
RECYCLE. RENEW. REGROW.
12 HOW TO KNOW COMPOST IS READY
Texture: dark, crumbly, soil-like
Smell: earthy, not sour or rotten.
Time: small home bin — 8-12 weeks
13 FAQ
Q: How often do I empty the bin?
A: Depends on usage — when finished compost accumulates (~8-12 weeks) or when bucket fills to ¾.
Q: Can I keep it indoors?
A: Yes — keep it under the sink or in a ventilated corner; ensure lid is sealed and odors controlled.
14 TROUBLESHOOTING (QUICK FIXES)
Bad rotten smell → Too wet / anaerobic. Add dry leaves, stir, open vents.
No activity / slow decomposition → Add a spoon of starter microbes or a small amount of finished compost; check moisture (should be like a damp sponge).
Fruit flies → Cover scraps, maintain mesh, add citrus in small amounts only.
Leaking tap → Re-seal with waterproof sealant or tighten.
Sustainable Living
Be part of the growing green movement.
Saves Money
Cut down fertilizer purchases.
Compact & Stylish
Fits easily in balconies, terraces, or backyards.

Skills Used

I created my product by using a used industrial paint or glue bucket/can as the base and modifying it to match the product the Design Specifications describe. To accomplish the final product, I required many miscellaneous materials such as charcoal, a mesh bag, a tap, a mesh disc and some kind of contraption to slightly close the vents, among other items that can be gathered from hardware stores. Implementing these elements into the product in the given time took **basic crafting skills** and **time-management skills**, however evaluating if to use them or not or which versions of said items are ideal required **judgement and analysis skills**. **All above mentioned skills, I possessed**. Though most of these things I can do myself with minimal help, for certain processes, such as the making of the charcoal filter, I needed to pass the task on to an **external professional with expertise in the required skills**, such as precise sewing skills. In conclusion, in order to create this product, I required **a combination of personal skills and external support**.

Skills	Usage
Basic crafting skills	➢ Cutting out vents and tap slot ➢ Attaching vent covers and tap ➢ Painting ➢ Applying stickers
Time-management skills	➢ Balancing the various simultaneously occurring manufacturing and testing steps.
Judgement and analysis skills	➢ Designing schedule ➢ Designing process ➢ Selecting appropriate materials and equipment

Changes to the Specifications

Specification	Change	Reason
Carbon steel stand	Wheeled plastic base	After developing the specifications, while contemplating how the compost bin, once full and heavy, would be made to move easily. Even the handle, while helping with gripping the bin, would not completely eliminate the effort of carrying, pushing or dragging the bin to move it. Upon realising that wheels could be added on the bottom of the bin I began hunting for some kind of wheeled contraption that I could attach to the product, significantly moving the friction caused by the weight while pushing or dragging it. I eventually came across a wheeled tray designed to be kept below heavy plant pots. I explored several similar products, eventually finding one that would fit the dimensions of my product. This new wheeled stand would help the product be easily moveable from one

		point to the other despite its weight.
Vents all over the Bin	A single 6 cm diameter hole on each side of the Bin covered by a plastic lid with tiny holes.	Holes all over the bin may cause compost material to fall out easily. However, if they were completely missing or too small, there would be no aeration. Therefore I decided to pick the middle ground where I would have a large hole on each of the four sides on the middle level of the bin. To prevent compost from falling out, I attached a cover. These covers have small holes all over them but protrude slightly outward from the hole in the Bin, preventing any compost that falls out from reaching all the way to the outer holes. This preserves the aeration factor while making sure the compost does not fall out. I have also made sure to make the holes on the slightly higher side since the liquid content, which would smell more, usually gathers in the lower portion.

Changes to the Process		
Step	**Change**	**Reason**
Painting the Bin as per the sketch with oil paints.	Applying a sticker showing an abundant garden.	The visual aesthetic was something I kept vague since the beginning as my primary concern was the functionality of the product. Hence I simply added a step to the process of painting the product as shown in the sketches, which was a brown and green colour scheme that would be thematic to composting and by extension gardening. Later on, in order to make the product stand out visually, I decided to use a paper sticker showing an abundant garden, also appropriate for the product since it would inspire the user to use the product to put together a similarly abundant garden. I did however add a layer of spray paint on the Bin and its parts, such as the tap, vent covers and the compost tea strainer holders.
Drilling vents	Cutting out vents with a heated knife.	Rather than drilling the vents via a drill, I decided to cut them out using a heated knife. This was simply because using a heated knife would

		be easier than using the drill, especially considering the change from smaller to larger vents.

Manufacturing Process Journal				
Sped-Up Video of the Manufacturing of the Re-Gen Bin				
Step Number	**Step:**	**Resources:**	**Approximate Estimation of Time taken:**	**Quality Checks (Success Criteria):**
1	Wash and dry the used plastic can that will act as the base of the product.	➢ Water ➢ Soap ➢ Cloth	10 minutes	The surface must be clean and dry and should not have moisture.
2	Use a heated knife to cut out holes in the bin, one being cut 12 cm below the top of the bucket with the diameter of 4.5 cm, two perpendicular to it on the same height and of the same size on both sides and one directly across it on the other side of the bucket so that there are now four evenly spaced vents.	➢ Drill ➢ Ruler	15 minutes	The holes must be smooth and evenly spaced.
	   			
3	Crush 1 kg of charcoal into small pieces and fill it into a bag with approximately the same dimensions. Once the bag is stitched close, attach it to the bottom of the lid.	➢ 1 kg Charcoal ➢ Mortar and Pestle ➢ Mesh bag ➢ Glue ➢ Needle ➢ Thread	1 day	The holes in the mesh bag must be small enough to not let the charcoal pieces fall out.

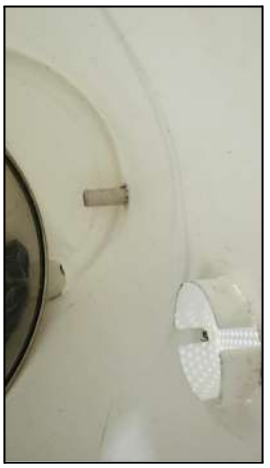


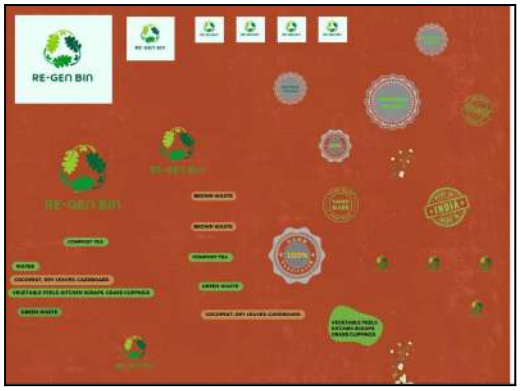



4	Take a wire filter disc the same diameter as the point in the bin that is 4 cm above its lowest point and fix it onto said point by attaching 4 small holders on the inner base surface using glue and resting the disc on it.	➢ Wire filter disc ➢ Glue	5 minutes	The holes in the wire filter disc should be sized so that the compost-tea can escape into the layer below it without the solid compost passing through.





5	Cut a hole with the diameter of 2 cm at the point directly below where the wire filter disc was attached and tightly fit the tap into it. Seal the area around it with water-proof sealant and let it dry.	➢ Tap ➢ Drill ➢ Water-proof sealant	20 minutes	The tap must open and close smoothly and without any leaks.
   				
6	Fix the bin onto the wheeled stand.	➢ Wheeled stand	No time	The Bin must fit on the stand and there should be enough clearance to fit the compost tea measuring cup.
7	Painting parts	➢ Brown oil spray paint	3 hours	The paint is even and dry.
   				

8	Design, print and apply all the stickers to the Bin. Cut the stickers that block any of the vents or the tap attachment slot.	➢ Canva	1 hour	Each sticker must be aligned with all the things it needs to be aligned with.
	       			
9	Test the final product based on the success criteria defined in the Design Specifications. If the product at this stage does not meet any particular specification or criteria, the relevant aspect of the product will be reworked.	Resources mentioned in Design Specifications.	Throughout the Process, 7 Weeks	All specifications must be successful.

Section 4: EVALUATING:

User Trial

The Re-Gen Bin was given to a user, Mrs Mangala Choudhary for 10 days, who composts often and fits the target audience of the Indian household. Upon using the product for 10 days, I interviewed the user about their experience. As I had only 10 days' time to have the product tested, I deemed it necessary to give it to someone I would consider an expert as she would be able to judge my product based on her past experience with composting.












Question	Responses
How many people are in your household?	There are 3 people in the household.
Where do you usually keep the Re-Gen Bin	I kept the Re-Gen Bin in the kitchen or backyard.
Was it easy to understand how to use the Re-Gen Bin Bin in the beginning?	Yes. The manual is a bit lengthy, though it does give you an understanding of using the product once you read it. Maybe there should be a video tutorial available online for the same.

Were there any steps throughout the 10 days of usage that were difficult?	Opening the lid was difficult as it was very tight. There should be something to make this easier.
How often did you add waste to the Re-Gen Bin?	I added fruit and vegetable peels everyday along with their necessary amount of brown waste according to the measuring cups.
How often do you stir the compost material?	Everyday, as I added compost material daily.
Is the Re-Gen Bin sufficient for your household waste?	Even in 10 days, only a little more than half of the Re-Gen Bin was filled.
Have you experienced any bad smells while using the Re-Gen Bin?	Much lesser than other compost bins I have used. Usually, whenever a compost bin is opened after many days, it gives out a blast of bad smell that stays for a long time. Here, the charcoal bag seemed to prevent such
Does the ventilation and charcoal bag system work to reduce odor while allowing aeration?	The compost came out well (meaning aeration was sufficient) and there was no bad smell either. So, yes, the system worked.
How did the included wheeled stand and supplementary composting equipment and material (gloves, claw, cocopeat) make the composting process easier?	Since I already do gardening often and composting occasionally, I already had gardening equipment like gloves, claw and cocopeat with me. Though, the measuring cups were a unique inclusion and made the calculation of how much brown and green waste to put much quicker. The wheeled stand made it easier to move the compost bin, especially when it was more than halfway full and quite heavy.
Were you able to produce compost successfully?	Though 10 days is not enough time for matter to compost, the compost was becoming thinner and smaller in volume, which is a sign of successful composting.
Is there something you would like to improve or change?	Yes. As I said before, the manual is very lengthy. So just have a summary of some of the important steps. Also, the green/brown waste measuring cups provided are too small and I had to fill the day's green waste in several rounds. So the green/brown waste measuring cups should be at least double their current size. Additionally, if you wish to include compost matter, you should also include ash and dry leaves, sawdust since these things are very important for compost.
The total price of this product will be ₹1550. Would you buy or recommend the product at this price?	Yes, since it includes a lot of different equipment and instructions, it will be a good deal. I have never seen this much included in composting kits that are even more expensive than this. So this will be a very good deal for beginners who don't have the knowledge or tools for composting. Though, I suggest that since you would also want to sell this to people who are experienced in composting and will already have a lot of the equipment and materials needed, you should sell a version without some of the extra material, which would be cheaper.

Testing Design Specifications			
Key: ✅Success 🟨Moderate Success ❌Failure			
Specification	**Testing Method**	**Success**	**Evidence**
Approx. 20-30L x 20-30B x 30-40H centimetres	The product being the accurate size as described in specifications.	✅	The measurement can be confirmed upon measuring the Re-Gen Bin. This can also be confirmed via the amazon listing of the bucket used as the base of the Re-Gen Bin, which displays the dimensions as 40H x 30L x 30B centimetres.
15-20 litre capacity	The product having a capacity accurate to the one described in the specifications.	✅	Volume of cylinder: pi*radius*2*height pi*13*2*40 = 21237.17 cubic centimeters 1 cubic centimeter = 0.001 litre 21237.17 cubic centimeters = 21.23 liters Therefore, yes the Re-Gen Bin can hold 15-20 litres of matter. This can also be confirmed via the amazon listing of the bucket used as the base of the Re-Gen Bin, which displays the capacity as 20 litres..
Plastic structure	Agreement of price among users (interviewees.)	✅	Rounded Pricing: ➢ Base bucket: ₹100 ➢ Ventilation mesh: ₹20 ➢ Tap: ₹80 ➢ Stand: ₹250 ➢ Charcoal bag: ₹60 ➢ Compost starter material: ₹100 ➢ Gardening claw and gloves: 180 ➢ Measuring cups: ₹100 ➢ Mesh strainer: ₹80 ➢ Paint: ₹200 ➢ Stickering and manual: ₹180 ➢ Final assembly (theoretical): ₹200 Total cost: ₹1550 The user approves of the total price and suggests that I sell a version of the product without some of the supplementary equipment that many regular gardeners may already have. Cost without supplementary compost equipment and materials (gloves, claw, cocopeat and microbes): ₹1270 Comparison of the price with the 4 best existing compost bins found during the initial research: The 4 were priced at ₹9719, ₹11560.40, ₹3500 and

			₹999, respectively. My product, with or without the additional gardening equipment, is much smaller than ¾ of them. The one that cost ₹999, the Kibiga Household Compost Bin, includes nothing other than the main bin, a bucket with vents and is made of metal. My product has much more functionality than it, including a compost-tea access method, measuring cups and more. Hence, my product being more expensive is not a bad thing and is completely justified.
Strong handle	Handle holding the weight of the product while full for 10 seconds without anything breaking.	✅	The handle is able to hold the weight of the filled bin for much more than 10 seconds.
Stirring claw and handgloves	Agreement of users (interviewees) that this makes the product easier to use.	✅	The users all said that the inclusion of this equipment will make it more helpful for beginners. To further prove the relevance of these inclusions, it is important to protect the hands from dirt, germs or irritants that may be harmful in any way. Hence gloves and gardening claws are used in composting so that you can add compost material and stir the compost while maintaining hygiene. A beginner may not already have any of these materials, and so it is logical to include these things with the product. (Blaskovich)
Supplementary compost matter	Agreement of users (interviewees) that this makes the product easier to use.	✅	The user said that the inclusion of this equipment will make it more helpful for beginners. Cocopeat, that is brown waste, needs to be 2.5 times the amount of green waste in the compost. While green waste can be kitchen waste, brown waste is not something users are likely to have at home. Therefore, it is logical to include supplementary compost matter with the product.
Wheeled stand	The stand being able to hold the compost bin at full capacity for 3 days, being stable with the compost bin kept on it and allowing for enough clearance to keep the compost-tea cup below the tap.	✅	Successful, as the stand has been continuously holding the Bin for the past 3 weeks without any damage being caused. Furthermore, the stand is tall enough for the compost tea measuring cup to fit below the tap as evident in the picture. 
Odor-controlling activated charcoal	Agreement of users (interviewees) that the	✅	The user states that the usual blast of foul odour that occurs when opening a compost bin to add compost

bag	Re-Gen Bin releases less foul odor than other compost bins they have with in the past.		material does not occur while opening this bin, in which the charcoal bag is kept on top of the compost.
Measuring cups	Agreement of users (interviewees) that the measuring cup system makes the process of composting easier for them to manage than it was before and the ratios being according to credible sources online.	🟨	While the ratios I used are accurate to several credible sources (United States Environmental Protection Agency), the user said that the cups are too small as sometimes, when more kitchen waste is produced, they would have to fill the bin via the cups in several rounds, which may become very inconvenient, with the cups needing to be at least double their current size.
Compost-tea access tap	The smooth opening and closing of the tap and its ability to release all stored compost-tea without leaking.	✅	The tap opens and closes smoothly, being able to release the liquid stored inside the bin without leakage, as evident in this video.
Water-dilution measurement	The compost-tea and water measuring cups being of the accurate size ratio in comparison with one another and the compost-tea cup being able to hold all compost tea released by the tap in 2 fills.	✅	The compost-tea measuring cup is able to hold the amount of compost tea released by the tap, as evident by the image below. 
Ventilation	The compost being in its ideal state, that is with a black-brown colour, wet and crumbly texture, earthly smell and 70-80% of the volume of the matter converted to compost by the end of 4 months. (Environmental Protection Agency)	✅	While a 10-day testing period was too small, the user stated that the volume of the compost was visibly decreasing with the matter becoming thinner, which is a sign of decomposition.
The above tests show that 11/12 of the sections of specifications have fulfilled success criteria, with 1 having only one aspect not fulfilled. Hence, on the basis of the specification tests, the Re-Gen Bin can be awarded 11.5/12 points.			

The Improved Product	
Improvements	**Justification**
Larger green/brown waste measuring cups. The green waste cup should be 200 ml and the brown waste cup should be 500 ml.	These new sizes preserve the 1:2.5 ratio and incorporate the feedback received from the user of at least doubling the size of the green/brown waste measuring cups.

The instruction manual should have a QR code linking to a video version of the instructions.	The user stated that the instruction manual is extremely lengthy. However, the detailed instructions are necessary for the proper usage of the product. Hence in order to preserve the detailed instructions while making the instructions less of a task to know, the instruction manual should have a QR code that leads to a video that shows the instructions, so that both users who are comfortable with reading the instructions and those who require a video can be satisfied.

Impact of the Re-Gen Bin

Comparison of the completed Re-Gen Bin with the initial Design Brief.

The final Re-Gen Bin matches very well with the original idea proposed in the design brief, a compact compost bin that would make the process of composting less challenging and would have a feature to balance odor-control with aeration. As evident by the final tests, the Re-Gen Bin matches all of these criteria.

Impact of the Re-Gen Bin on the Target Audience

The target audience, an Indian Household, is satisfied with the product. Since the initial conception of the product, every aspect has been made to make sure it solves the problem it is meant to solve, from the creation of Design Specifications to its final making and the changes made to the design, it has been made sure that the Re-Gen Bin is solving the problem it was made to solve, which guaranteed its success from the beginning.

Giving Rise to New Problems

The Re-Gen Bin, while its merits cannot be ignored, could impact human habits in a way that could backfire. Perhaps, composting made more accessible could encourage people to waste more food. If food is wasted on extremely large scales could cause severe environmental impact such as greenhouse gas emissions. (Lewis) If the habit grows faster than composting and composting technology can grow in the future, it will do massive harm to the environment.

Is the problem solved?

Yes, the problem statement was ‘Many Indian households do not practice composting, which increases the amount of food waste produced.’ As evident by the adherence to the design specifications which were all backed by credible sources and the success of all of the tests that were based on the specifications, this problem can be deemed resolved by my product.
However we must make it clear that the problem statement that was selected was only a small part of a much larger problem. If we go back to why this problem was chosen, it was to build a sustainable world. Of course, a single product cannot build a sustainable world. The Re-Gen Bin will be just a part of a whole ecosystem of products that will contribute to the cause of a sustainable world with solutions to all problems. And those are the kinds of products that the Design Cycle encourages us to create.



Thank You

Sources
Inquiring and Analysing

Works Cited

Agarwal, Shandar. “India Food Waste Crisis: Current Realities and Future Challenges - Naas India.” *Naas India*, 7 Nov. 2024, naasindia.org/india-food-waste-crisis/?utm_source=chatgpt.com.

Aktar, Wasim, et al. “Impact of Pesticides Use in Agriculture: Their Benefits and Hazards.” *Interdisciplinary Toxicology*, vol. 2, no. 1, Mar. 2009, pp. 1–12, https://doi.org/10.2478/v10102-009-0001-7. National Library of Medicine.

Arora, Sanjoli. “The Big Food Waste Problem in India: Causes, Concerns & Actions.” *Conscious Charcha*, 18 June 2023, consciouscharcha.com/food-waste-india-problem/.

Blaskovich, Mark. “(at Least) Five Reasons You Should Wear Gardening Gloves.” *Imb.uq.edu.au*, 1515, imb.uq.edu.au/article/2018/01/least-five-reasons-you-should-wear-gardening-gloves.

Climate and Clean Air Coalition. “Methane | Climate & Clean Air Coalition.” *Www.ccacoalition.org*, 2021, www.ccacoalition.org/short-lived-climate-pollutants/methane.

Ministry of Environment. “Waste Generation in the Country.” *Pib.gov.in*, 2024, www.pib.gov.in/PressReleaseIframePage.aspx?PRID=2003989.

Mohan, Vishwa. “74 Million Tonnes of Food Amounting to 22% of Foodgrain Output Wasted in India Every Year.” *The Times of India*, 1 Nov. 2023,

timesofindia.indiatimes.com/india/74-million-tonnes-of-food-amounting-to-22-of-foodgrain-output-wasted-in-india-every-year/articleshow/104875206.cms.

PIB. “Department of Agriculture and Farmers’ Welfare Releases Final Estimates of Major Agricultural Crops for 2023-24.” *Pib.gov.in*, 2023, www.pib.gov.in/PressReleasePage.aspx?PRID=2058534.

US EPA. “Approaches to Composting.” *Www.epa.gov*, 19 Aug. 2015, www.epa.gov/sustainable-management-food/approaches-composting.

---. “Basic Information about Landfill Gas .” *US EPA*, United States Environmental Protection Agency, 20 Sept. 2024, www.epa.gov/lmop/basic-information-about-landfill-gas.

Developing Ideas, Creating the Solution, Evaluating

These sources are shown together as the information seen in parts 3 and 4 are based on the research from part 2.

Works Cited

Atlas Fibre. “Advantages of Plastics Compared to Other Materials.” *Atlas Fibre*, 10 November 2023, https://www.atlasfibre.com/advantages-of-plastics/. Accessed October 2025.

Brikov, Michail, et al. “Increasing the Strength and Impact Toughness of Carbon Steel Using a Nanosized Eutectoid Resulting from Time-Controlled Quenching.” *Advanced High-Strength Steels (AHSS) Prepared as Bulk and Powder Metal*, 2024. https://www.mdpi.com/1996-1944/17/15/3696#:~:text=The%20strength%20of%20steels%20can,10%E2%80%93100%20nm%2C%20respectively. Accessed October 2025.

Environmental Protection Agency. “Composting At Home | US EPA.” *EPA*, 20 August 2025, https://www.epa.gov/recycle/composting-home. Accessed October 2025.

Farming and Agriculture Organization of USA. “Chapter 11 - Briquetting of charcoal.” *Food and Agriculture Organization of the United Nations*, https://www.fao.org/4/x5328e/x5328e0c.htm. Accessed October 2025.

kloprogge, j. “Clays and the Origin of Life: The Experiments.” *PubMed Central*, https://pmc.ncbi.nlm.nih.gov/articles/PMC8880559/. Accessed 8 October 2025.

Mohan, Aishvarya V. “Activated Carbon.” *International Journal of Innovative Science and Research Technology*, vol. 5, no. 4, 2020, 829, 830. https://www.ijisrt.com/assets/upload/files/IJISRT20APR790.pdf. Accessed October 2025.

New Berlin Plastics. “The Biggest Reasons to Replace Metals with Plastic.” *nbplastics.com*, 15 02 2018, https://nbplastics.com/resources/biggest-reasons-replace-metals-plastic/#:~:text=Using%20Plastics%20to%20Cut%20Manufacturing,this%20can%20reduce%20maintenance%20costs. Accessed October 2025.

Roebuck, Anthony. “How Much Does Compost (Really) Weigh?” *helpmecompost.com*, nodexis.co, 2025, https://helpmecompost.com/compost/basics/how-much-does-compost-weigh/. Accessed October 2025.

Lewis, Jangira. “How Does Food Waste Affect the Environment?” *Earth.org*, 17 Oct. 2022, earth.org/how-does-food-waste-affect-the-environment/.

RSPinc. “Metal and Plastic: What are the Differences? | RSP.” *RSP Inc.*, 2 April 2024, https://www.rspinc.com/blog/plastic-injection-molding/plastic-vs-metal/. Accessed October 2025.

The Welding Institute. “What is the Difference Between Carbon Steel and Stainless Steel?” *TWI*, https://www.twi-global.com/technical-knowledge/faqs/carbon-steel-vs-stainless-steel. Accessed October 2025.

Yin, Jie, et al. “Compost tea: Preparation, utilization mechanisms, and agricultural applications potential – A comprehensive review.” *Environmental Technology & Innovation*, vol. 38, 2025, 1, 6. https://pdf.sciencedirectassets.com/308311/1-s2.0-S2352186425X0002X/1-s2.0-S2352186425001233/main.pdf?X-Amz-Security-Token=IQoJb3JpZ2luX2VjEOL%2F%2F%2F%2F%2F%2F%2F%2F%2F%2FwEaCXVzLWVhc3QtMSJIMEYCIQDFfltiP4pYiSGPE1ISIIASLDLAuoOwm%2FHL6QhRYyt7OQIhALwT4cdUowm. Accessed 10 2025.

Product Listings (Amazon)

- **Kibiga Kitchen Composting Bin**
- **Marcytop Outdoor Composter**
- **Daily Dump Prithvi Khamb Composter**
- **Trustbasket Indoor Compost Bin**
- **Velura Inground Worm Composter**
- **Dostbin Smart Automatic Composter**
- **Ez4garden Garden Leaf Waste Bag**
- **Waqif Kitchen Countertop Compost Bin**
- **AKXY Home Composting Machine**